Waqar Ali Soomro

MSc Applied Mathematics (Stochastics): Budapest University of Technology and Economics

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Date of birth: 3 January 1999 · Nationality: Pakistani

EDUCATION

Feb 2024 – Jul 2026 — **MSc in Applied Mathematics (specialisation: Stochastics)**
Budapest University of Technology and Economics (BME), Institute of Mathematics
Final classification: *Excellent with highest honours* (4.73 / 5.00), conferred 07 July 2026.
Master's thesis: Limiting Distribution of the Elephant Random Walk in the Superdiffusive Regime. *Focus areas:* probability theory, stochastic processes and analysis, martingales, limit theorems, and mathematical statistics.

From Sep 2026 — **Partial studies (post-MSc): BME Institute of Mathematics**
Advanced coursework in probability and stochastic processes in preparation for doctoral research, alongside continued teaching.

May 2022 – Jan 2024 — **M.S. in Applied Mathematics,coursework completed; degree not conferred**
NED University of Engineering and Technology, Karachi, Pakistan
Cumulative GPA 3.96 / 4.00 across completed coursework.

Oct 2017 – Dec 2021 — **BE in Civil Engineering**
Mehran University of Engineering and Technology, Campus Khairpur Mir's, Pakistan

RESEARCH EXPERIENCE

2026 — **Master's thesis, Elephant Random Walk in the superdiffusive regime**
BME · Supervisor: Dr Bálint Vető
Largely self-contained treatment of the two established routes to the superdiffusive limit: Bercu's martingale method and the Guérin–Laulin–Raschel fixed-point equation with proofs expanded and filled in. The 3rd abd 4th moment computations, only sketched in Bercu's paper, are carried out in full via a Gamma-telescoping argument, and the two approaches are shown to yield identical moments up to fourth order.

2024 – 2025 — **Project Laboratory, BME, Institute of Mathematics**
Panel-data econometric analysis of ten-year sovereign bond yields against debt-to-GDP ratios for the Visegrád Four, in R and Python, followed by machine-learning methods on structured graph data (supervised by Dr Roland Molontay).

PUBLICATIONS AND CONFERENCE CONTRIBUTIONS

2026 — M. Meghwar, **W. A. Soomro**, T. Kovács, B. Lal. "ACF-Guided LSTM Framework for Multi-Step Indoor Temperature Forecasting in IoT Smart Buildings." Accepted at *IEEE Smart World Congress 2026*, Italy; presentation scheduled for 7–11 September 2026.

7 Jul 2026 — **Master's thesis defence — BME, Institute of Mathematics**
Public defence of *Limiting Distribution of the Elephant Random Walk in the Superdiffusive Regime*; committee: Prof. Marianna Bolla, Dr Balázs Ráth, Dr Attila Nagy.

TEACHING EXPERIENCE

From Sep 2026 — **Demonstrator — Probability Theory 1**
Budapest University of Technology and Economics, Institute of Mathematics

From Sep 2026 — **Teaching assistant — Calculus 1**
Budapest University of Technology and Economics, Institute of Mathematics

Feb 2026 – Jun 2026 — **Teaching assistant — Calculus 2 (groups G2f and A2a)**
Budapest University of Technology and Economics, Institute of Mathematics

LEADERSHIP AND SERVICE

Selected 2026	**Examination supervisor — International Mathematical Olympiad (IMO) 2027, Budapest** Selected by the national organising committee to supervise contest sessions of the 68th IMO, to be held in Budapest in July 2027.
Since Jun 2026	**Founder — BME International Mathematics Support Society (IMSS)** Student initiative supporting international students at BME with mathematics, to be officially inaugurated in the 2026/27 academic year under the BME International Affairs Committee (IAC).
2022	**Flood relief volunteer — northern Sindh, Pakistan** Organised fundraising with a group of friends and distributed rations and clothing to families displaced by the 2022 floods.
2016 – 2017	**Volunteer teacher — Lakhi, Sindh** Taught reading and basic mathematics to primary-age children in the village over approximately eighteen months before leaving for university.

ADDITIONAL MATHEMATICAL TRAINING

Supplementary self-directed study undertaken alongside the M.S. and MSc programmes, in transition from an engineering to a mathematics background.

2022 – 2023	*MITx (MIT):* 6.431x Probability: The Science of Uncertainty and Data; 18.6501x Fundamentals of Statistics; 18.032x, 18.033x, 18.03Fx Differential Equations series (2×2 systems; linear algebra and $N\times N$ systems; Fourier series and PDEs).
2020 – 2023	*Other:* Operations Research I–III (National Taiwan University, verified); Matrix Algebra for Engineers; Vector Calculus for Engineers (Hong Kong University of Science and Technology, verified).

SCHOLARSHIPS AND AWARDS

2024 – 2026	**Stipendium Hungaricum Scholarship**: full Hungarian government scholarship for MSc study at BME, administered by Tempus Public Foundation.
2023	**Prime Minister's National Laptop Scheme, Pakistan**: awarded on academic merit by the Higher Education Commission (HEC).
2017 – 2021	**Government of Pakistan Need based Scholarship**: undergraduate study at Mehran University of Engineering and Technology.

SKILLS AND LANGUAGES

Mathematical	Probability theory, stochastic processes, martingales and limit theorems, Monte Carlo and stochastic simulation, statistical inference; time-series and panel-data econometrics, optimisation, machine learning.
Computational	Python (NumPy, SciPy, pandas, statsmodels, Matplotlib), R, MATLAB, LaTeX; working knowledge of C++ and Git.
Languages	Sindhi (native) · Urdu (native) · English (B2, onSET, Universität Szeged).

REFEREES

Dr Bálint Vető Associate Professor, Department of Stochastics, BME Institute of Mathematics; MSc thesis supervisor. vetob@math.bme.hu

Prof. Dr Andreas Eberle Professor, Institut für Angewandte Mathematik (Probability Theory and Stochastic Analysis), Universität Bonn; prospective doctoral supervisor. eberle@uni-bonn.de